

Micro-Exercise — The 3-Sentence Abstract

Mastering the Scientific Narrative · 4 minutes of writing, a lifetime of reuse

Pick ONE scenario card (page 2), then write your abstract below — numbers may be invented but must be plausible.

Name / team: _____

Scenario chosen: _____

1 CONTEXT / PROBLEM — set the stakes, name the barrier

Where? Who is affected? Which numbers make it urgent? What blocks today's solutions?

e.g. "In African agricultural zones, crop disease X causes major yield losses, but existing vision models require high-compute resources."

2 METHOD — your one-sentence answer

"We introduce [Name], a [key property] [system] designed for [constraint]." One system, one property, one constraint.

e.g. "We introduce AgroNet, an offline, quantized network optimized for low-end mobile devices."

3 IMPACT / RESULTS — numbers, not adjectives

Which data? Which metric? How much better than which baseline? What does it enable?

e.g. "Our approach reaches 78% accuracy on local data while cutting inference time by 40% versus baselines."

SELF-CHECK — tick before reading it aloud

- ☐ A stranger could FUND this project after my 3 sentences.
- ☐ Sentence 3 contains at least one NUMBER.
- ☐ The gap (what blocks the world today) is explicit.
- ☐ I named the region / context — my African anchor is a strength, not a footnote.
- ☐ Zero empty adjectives ("good", "promising", "novel") survive.

Scenario Cards — pick one, or invent your own

Print & cut along the dashed lines · facts below are realistic starting points, adapt them freely

A · CLIMATE

Flood forecasting for West African cities

THE SITUATION

Seasonal floods displace hundreds of thousands each year in coastal and riverine cities (e.g., Lagos, Niamey, Conakry). Station coverage is sparse; satellite data exist but heavy models cannot run locally.

THINK ABOUT

- Who issues the warning, and how early?
- Fuse rainfall, river gauges, satellite?
- Metrics: lead time (h), hit / false-alarm rate.

Your mission: 3 sentences — Context+Gap → Method → Numbers.

B · AGRICULTURE

Pest & disease detection for smallholders

THE SITUATION

Smallholders grow most of the continent's food; fall armyworm can destroy a maize season. Agronomists are rare; smartphones are common but low-end, and connectivity is intermittent in rural areas.

THINK ABOUT

- Offline or SMS-based? Which crops, region?
- Labeled images: farmers? extension officers?
- Metrics: accuracy, on-device latency, yield.

Your mission: 3 sentences — Context+Gap → Method → Numbers.

C · HEALTH

Malaria diagnosis support in rural clinics

THE SITUATION

Manual microscopy takes 30+ min per slide; expert microscopists are scarce outside big hospitals. Misdiagnosis drives untreated cases and drug misuse. Clinics have a basic microscope and one Android phone.

THINK ABOUT

- Phone camera on microscope? Which stages?
- Ethics: consent, data protection, validation.
- Metrics: sensitivity vs expert, sec per slide.

Your mission: 3 sentences — Context+Gap → Method → Numbers.

D · LANGUAGE

NLP for African local languages

THE SITUATION

Thousands of African languages are digitally under-resourced — few corpora, fewer benchmarks (Susu, Pular, Hausa, Yorùbá...). Speech/text services for health or agri advice fail for these speakers.

THINK ABOUT

- Task: translation, ASR, advice chatbot?
- Data: community, radio, participatory?
- Metrics: BLEU/WER, human eval, corpus hrs.

Your mission: 3 sentences — Context+Gap → Method → Numbers.